Modeling Climate Driven *Ambystoma tigrinum* Virus Dynamics:

Integrating Temperature, Landscape Fragmentation, and Host Behavior using epymorph

Andrew Kerfoot, Kelsey Banister, Joseph Mihaljevic

School of Informatics, Computing, and Cyber Systems and The Pathogen and Microbiome Institute, Northern Arizona University, Flagstaff, AZ 86011

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Acknowledgements

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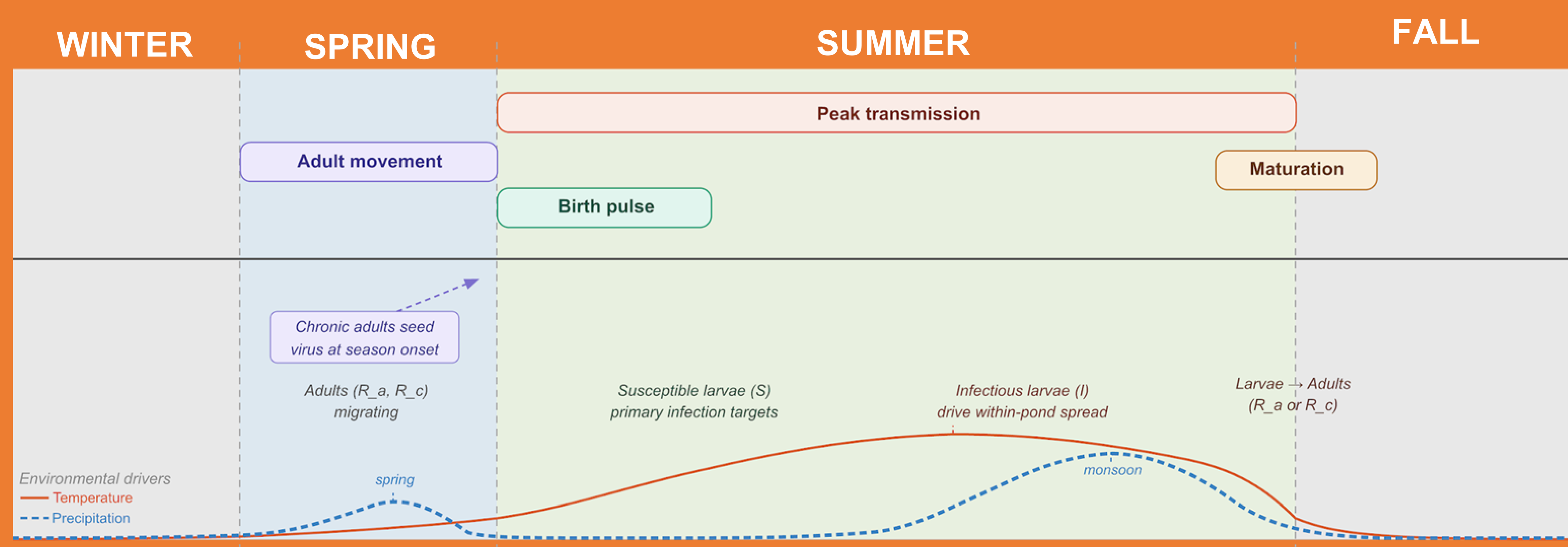
Introduction

- Climate change is reshaping wildlife disease dynamics by altering pathogen transmission, host susceptibility, and outbreak timing.
- ***Ambystoma tigrinum*** **virus (ATV)** causes severe mortality in Arizona tiger salamander larvae, with pond-wide epizootics capable of collapsing seasonal cohorts.
- ATV transmission may be amplified by warming temperatures, fragmented habitat networks, and adult dispersal between breeding ponds.
- The combined effects of **climate, landscape structure,** and **host movement** remain poorly understood.
- We developed a spatially explicit metapopulation model to test how these interacting processes shape ATV outbreaks.

Discussion

- Spatial structure and pond **connectivity** strongly influence ATV outbreak timing and severity.
- Adult **movement** linked local ponds, allowing isolated outbreaks to spread across the landscape.
- Uniform **temperature** forcing was incorporated in Model 3; comparative climate scenarios remain a future direction.
- Preliminary simulations suggest temperature-dependent transmission may alter epidemic timing and intensity.
- Spatial modelling offers a useful tool for forecasting future risk and supporting amphibian conservation.



Methods

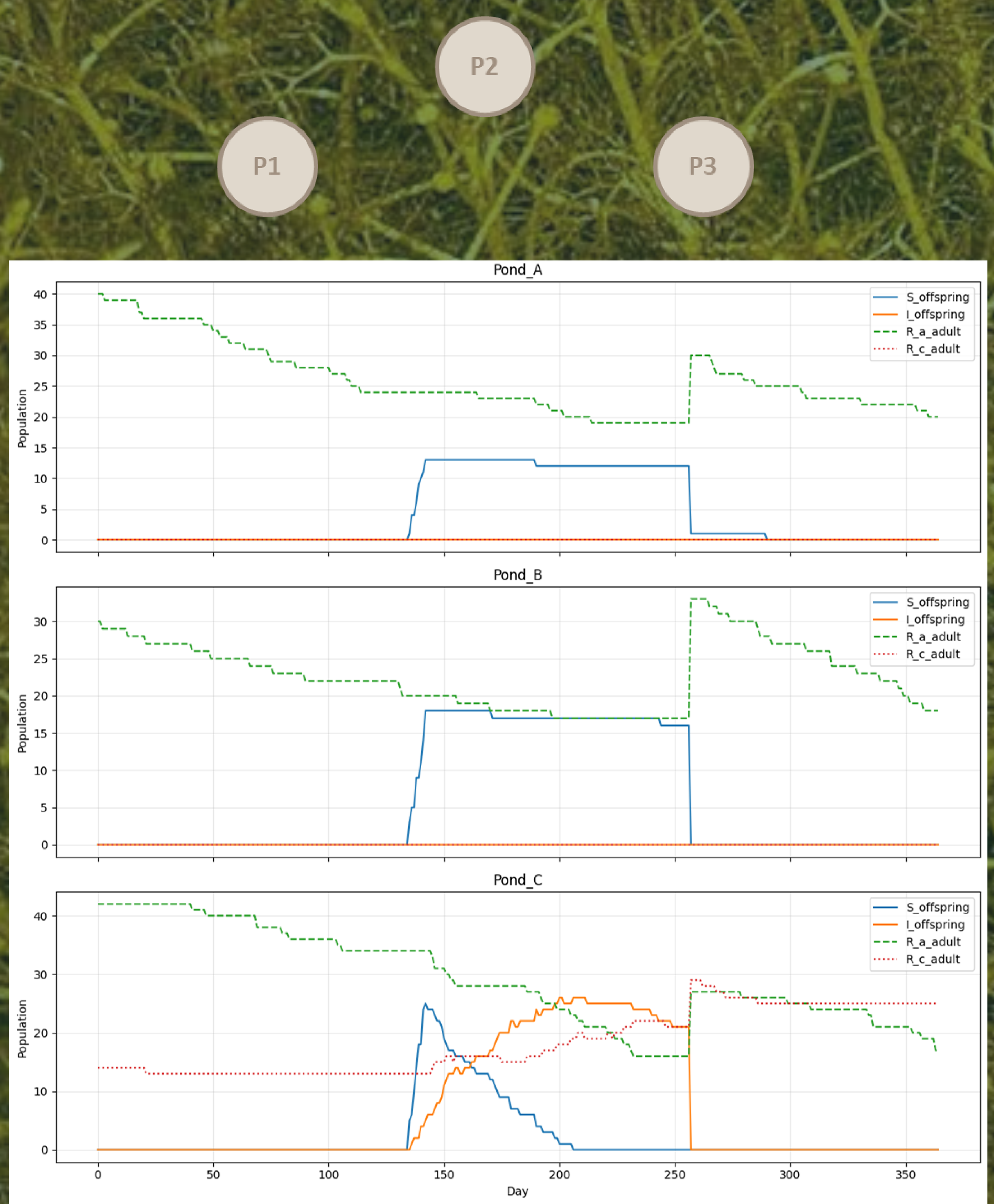
Custom **three-pond metapopulation model** built in Python using epymorph 1.1.0.

- Four epidemiological classes per pond:
  - S = Susceptible larvae
  - I = Infectious larvae
  - R_a = Cleared adults
  - R_c = Chronic adults
- Larvae drive local transmission; adults move seasonally between ponds

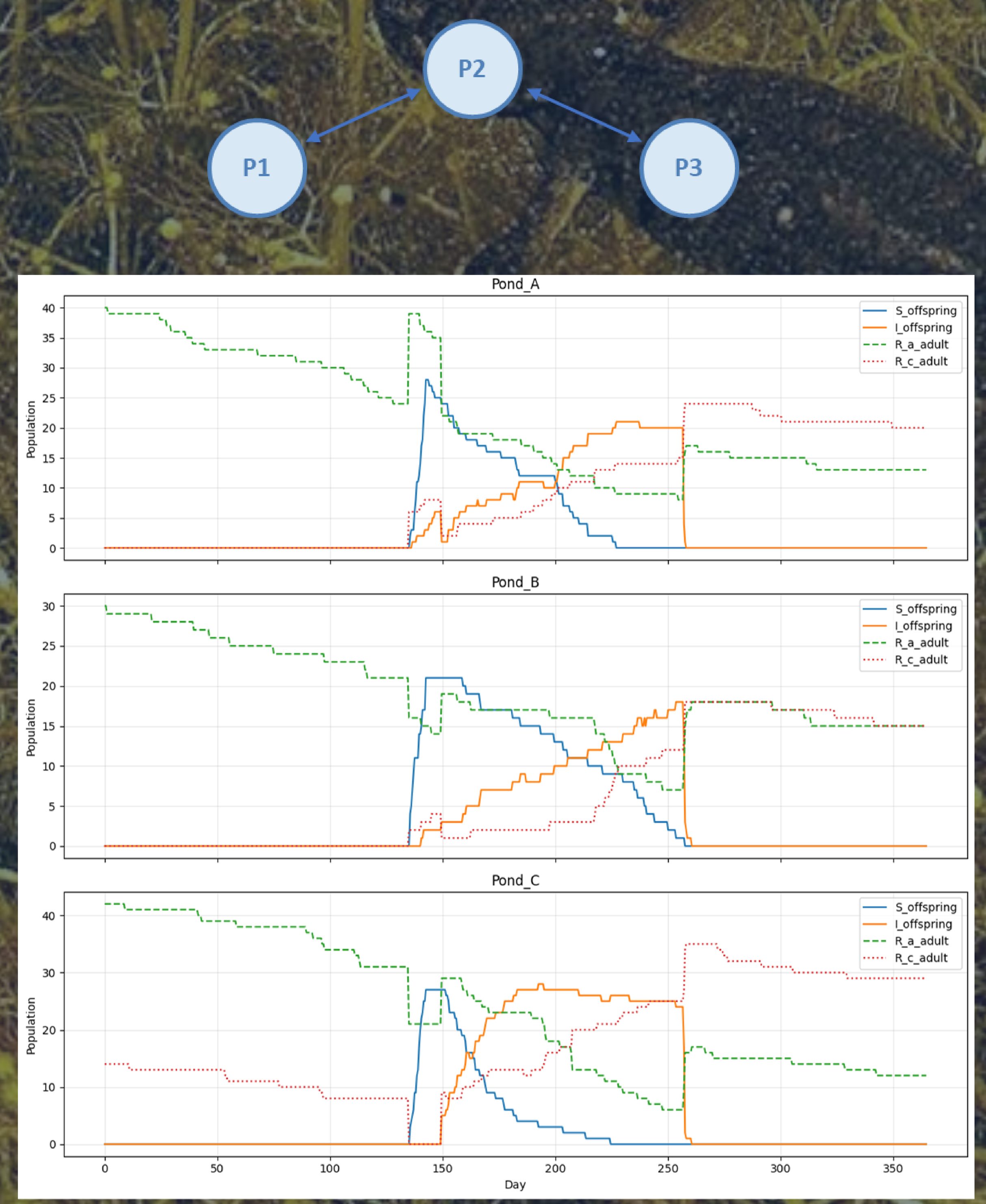
Sequential model development:

- Model 1: No movement, no temperature
- Model 2: Seasonal adult movement
- Model 3: Temperature-dependent transmission and mortality

MODEL 1: NO MOVEMENT, NO TEMPERATURE



MODEL 2: MOVEMENT, NO TEMPERATURE



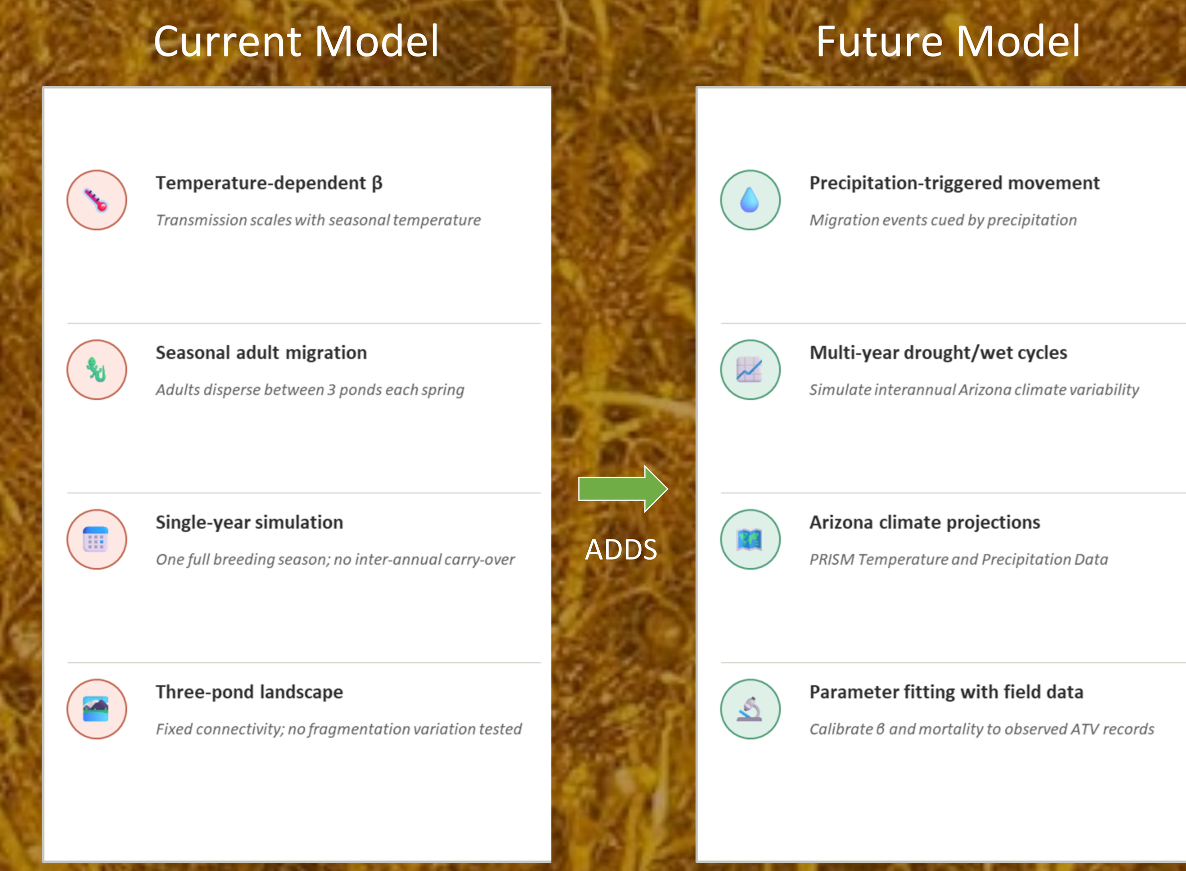
MODEL 3: MOVEMENT, TEMPERATURE

*All ponds simulated under the same temperature regime

FUTURE MODEL 4: MOVEMENT, TEMPERATURE, PRECIPITATION, MULTI-YEAR

Precipitation-triggered movement, Arizona climate projections, parameter fitting with field data.

*Additional temperature scenario tests available via QR code



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Scan for additional temperature sensitivity tests, references, and supplemental figures: